

Data Base Management System (DBMS)

2/09/2026

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Roll No : 15

Batch : B1

Assignment No 3 – SQL Operators

Question 1: Employee Management

Table Structure:

EmployeeDetails(EmpID, EmpName, MonthlySalary, City)

Perform the following SQL queries

Queries:

- a) Display the EmpID, EmpName, and MonthlySalary of all employees.

```
mysql> select EmpID,EmpName,Monthllysalary from EmployeeDetails;
+-----+-----+-----+
| EmpID | EmpName      | Monthllysalary |
+-----+-----+-----+
|      1 | Harshad Teli |          60000 |
|      2 | Anurag Lad   |          45000 |
|      3 | Vardhan Patki |          35000 |
|      4 | Kartika Phalle |          75000 |
|      5 | Anjali Patil  |          25000 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> █
```

b) Display employees whose MonthlySalary is greater than **30000**.

```
mysql> select * from EmployeeDetails where MonthlySalary > 30000;
+-----+-----+-----+-----+-----+
| EmpID | EmpName          | Department | MonthlySalary | City   |
+-----+-----+-----+-----+-----+
| 1     | Harshad Teli     | IT         | 60000         | KOP    |
| 2     | Anurag Lad       | IT         | 45000         | Pune   |
| 3     | Vardhan Patki    | Marketing  | 35000         | Satara |
| 4     | Kartika Phalle   | IT         | 75000         | Karad  |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

c) Display employees whose MonthlySalary is less than or equal to **50000**.

```
mysql> select * from EmployeeDetails where MonthlySalary <= 50000;
+-----+-----+-----+-----+-----+
| EmpID | EmpName          | Department | MonthlySalary | City   |
+-----+-----+-----+-----+-----+
| 2     | Anurag Lad       | IT         | 45000         | Pune   |
| 3     | Vardhan Patki    | Marketing  | 35000         | Satara |
| 5     | Anjali Patil     | IT         | 25000         | Mumbai |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> █
```

d) Display employees who belong to the **IT department** and have a MonthlySalary greater than **30000**.

```
mysql> select * from EmployeeDetails where Department='IT' AND MonthlySalary > 30000;
+-----+-----+-----+-----+-----+
| EmpID | EmpName          | Department | MonthlySalary | City   |
+-----+-----+-----+-----+-----+
| 1     | Harshad Teli     | IT         | 60000         | KOP    |
| 2     | Anurag Lad       | IT         | 45000         | Pune   |
| 4     | Kartika Phalle   | IT         | 75000         | Karad  |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> █
```

e) Display employees who belong to either the **HR department** or **IT department**.

```
mysql> select * from EmployeeDetails where Department='IT' OR Department='HR';
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City   |
+-----+-----+-----+-----+-----+
|      1 | Harshad Teli | IT         |          60000 | KOP    |
|      2 | Anurag Lad   | IT         |          45000 | Pune   |
|      4 | Kartika Phalle | IT        |          75000 | Karad  |
|      5 | Anjali Patil | HR         |          25000 | Mumbai |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

f) IDisplay employees whose Department is **IT, HR, or Finance**.

```
mysql> select * from EmployeeDetails where Department IN ('IT','HR','Finance');
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City   |
+-----+-----+-----+-----+-----+
|      1 | Harshad Teli | IT         |          60000 | KOP    |
|      2 | Anurag Lad   | IT         |          45000 | Pune   |
|      4 | Kartika Phalle | Finance    |          75000 | Karad  |
|      5 | Anjali Patil | HR         |          25000 | Mumbai |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

g) Display employees whose MonthlySalary is between **25000 and 50000**.

```
mysql> select * from EmployeeDetails where MonthlySalary BETWEEN 25000 AND 50000;
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City   |
+-----+-----+-----+-----+-----+
|      2 | Anurag Lad   | IT         |          45000 | Pune   |
|      3 | Vardhan Patki | Marketing  |          35000 | Satara |
|      5 | Anjali Patil | HR         |          25000 | Mumbai |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

h) Display employees whose EmpName starts with the letter '**A**'.

```
mysql> select * from EmployeeDetails where EmpName LIKE 'A%';
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City   |
+-----+-----+-----+-----+-----+
|      2 | Anurag Lad   | IT         |          45000 | Pune   |
|      5 | Anjali Patil | HR         |          25000 | Mumbai |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
```

i) Display employees who are **not from the IT department**.

```
mysql> select * from EmployeeDetails where Department != 'IT';
+-----+-----+-----+-----+-----+
| EmpID | EmpName          | Department | MonthlySalary | City    |
+-----+-----+-----+-----+-----+
|      3 | Vardhan Patki    | Marketing  |      35000    | Satara  |
|      4 | Kartika Phalle   | Finance    |      75000    | Karad   |
|      5 | Anjali Patil     | HR         |      25000    | Mumbai  |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> select * from EmployeeDetails where NOT Department='IT';
+-----+-----+-----+-----+-----+
| EmpID | EmpName          | Department | MonthlySalary | City    |
+-----+-----+-----+-----+-----+
|      3 | Vardhan Patki    | Marketing  |      35000    | Satara  |
|      4 | Kartika Phalle   | Finance    |      75000    | Karad   |
|      5 | Anjali Patil     | HR         |      25000    | Mumbai  |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> 
```

j) Display EmpID, EmpName, MonthlySalary, and calculate the **Annual Salary**.

```
mysql> select EmpID,EmpName,MonthlySalary,(MonthlySalary*12) AS AnnualSalary from EmployeeDetails;
+-----+-----+-----+-----+
| EmpID | EmpName          | MonthlySalary | AnnualSalary |
+-----+-----+-----+-----+
|      1 | Harshad Teli     |      60000    |    720000    |
|      2 | Anurag Lad       |      45000    |    540000    |
|      3 | Vardhan Patki    |      35000    |    420000    |
|      4 | Kartika Phalle   |      75000    |    900000    |
|      5 | Anjali Patil     |      25000    |    300000    |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> 
```

Question 2: Student Management

Table Structure:

StudentMaster(StudentID, StudentName, Course, Age, Email)

Perform the following SQL operations:

Queries:

a) Display the StudentID, StudentName, and Course of all students.

```
mysql> select StudentID,StudentName,Course from StudentMaster;
+-----+-----+-----+
| StudentID | StudentName      | Course |
+-----+-----+-----+
|      1001 | Anadhut Phalle   | BCA    |
|      1002 | Shree Patil      | MCA    |
|      1003 | Shreya Vibhute   | MBA    |
|      1004 | Neha Sutar       | BSC    |
|      1005 | Anurag Lad       | BCA    |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> 
```

b) Display students whose StudentAge is greater than 20.

```
mysql> select * from StudentMaster where StudentAge > 20;
+-----+-----+-----+-----+-----+
| StudentID | StudentName      | Course | StudentAge | Email                |
+-----+-----+-----+-----+-----+
|      1002 | Shree Patil      | MCA    |      23    | shree@gmail.com     |
|      1003 | Shreya Vibhute   | MBA    |      21    | shreya@gmail.com    |
|      1005 | Anurag Lad       | BCA    |      21    | anurag@gmail.com    |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> 
```

c) Display students whose StudentAge is less than or equal to 25.

```
mysql> select * from StudentMaster where StudentAge <=25;
+-----+-----+-----+-----+-----+
| StudentID | StudentName      | Course | StudentAge | Email                |
+-----+-----+-----+-----+-----+
|      1001 | Anadhut Phalle   | BCA    |      20    | avadhut@gmail.com    |
|      1002 | Shree Patil      | MCA    |      23    | shree@gmail.com      |
|      1003 | Shreya Vibhute   | MBA    |      21    | shreya@gmail.com     |
|      1005 | Anurag Lad       | BCA    |      21    | anurag@gmail.com     |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

d) Display students who are studying **MCA** and whose StudentAge is greater than 20.

```
mysql> select * from StudentMaster where Course='MCA' AND StudentAge >20;
+-----+-----+-----+-----+-----+
| StudentID | StudentName      | Course | StudentAge | Email                |
+-----+-----+-----+-----+-----+
|      1002 | Shree Patil      | MCA    |      23    | shree@gmail.com      |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> █
```

e) Display students who are studying either **MCA or BCA**.

```
mysql> select * from StudentMaster where Course='MCA' OR Course='BCA';
+-----+-----+-----+-----+-----+
| StudentID | StudentName      | Course | StudentAge | Email                |
+-----+-----+-----+-----+-----+
|      1001 | Anadhut Phalle   | BCA    |      20    | avadhut@gmail.com    |
|      1002 | Shree Patil      | MCA    |      23    | shree@gmail.com      |
|      1005 | Anurag Lad       | BCA    |      21    | anurag@gmail.com     |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> █
```

f) Display students whose Course is **MCA, BCA, or MBA**.

```
mysql> select * from StudentMaster where Course IN ('MCA','BCA','MBA');
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | Email |
+-----+-----+-----+-----+-----+
| 1001 | Anadhut Phalle | BCA | 20 | avadhut@gmail.com |
| 1002 | Shree Patil | MCA | 23 | shree@gmail.com |
| 1003 | Shreya Vibhute | MBA | 21 | shreya@gmail.com |
| 1005 | Anurag Lad | BCA | 21 | anurag@gmail.com |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

g) Display students whose StudentAge is between **18 and 25**.

```
mysql> select * from StudentMaster where StudentAge BETWEEN 18 AND 25;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | Email |
+-----+-----+-----+-----+-----+
| 1001 | Anadhut Phalle | BCA | 20 | avadhut@gmail.com |
| 1002 | Shree Patil | MCA | 23 | shree@gmail.com |
| 1003 | Shreya Vibhute | MBA | 21 | shreya@gmail.com |
| 1005 | Anurag Lad | BCA | 21 | anurag@gmail.com |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> 
```

h) Display students whose StudentName starts with the letter **'S'**.

```
mysql> select * from StudentMaster where StudentName LIKE 'S%';
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | Email |
+-----+-----+-----+-----+-----+
| 1002 | Shree Patil | MCA | 23 | shree@gmail.com |
| 1003 | Shreya Vibhute | MBA | 21 | shreya@gmail.com |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
```


i) Display students who are **not studying MCA**.

```
mysql> select * from StudentMaster where Course!='MCA';
```

StudentID	StudentName	Course	StudentAge	Email
1001	Anadhut Phalle	BCA	20	avadhut@gmail.com
1003	Shreya Vibhute	MBA	21	shreya@gmail.com
1004	Neha Sutar	BSC	26	neha@gmail.com
1005	Anurag Lad	BCA	21	anurag@gmail.com

```
4 rows in set (0.00 sec)
```



```
mysql> select * from StudentMaster where NOT Course='MCA';
```

StudentID	StudentName	Course	StudentAge	Email
1001	Anadhut Phalle	BCA	20	avadhut@gmail.com
1003	Shreya Vibhute	MBA	21	shreya@gmail.com
1004	Neha Sutar	BSC	26	neha@gmail.com
1005	Anurag Lad	BCA	21	anurag@gmail.com

```
4 rows in set (0.00 sec)
```



```
mysql> 
```

j) Display students whose StudentName contains the letter 'a' and whose StudentAge is greater than 20.

```
mysql> select * from StudentMaster where StudentName LIKE "%a%" AND StudentAge>20;
```

StudentID	StudentName	Course	StudentAge	Email
1002	Shree Patil	MCA	23	shree@gmail.com
1003	Shreya Vibhute	MBA	21	shreya@gmail.com
1004	Neha Sutar	BSC	26	neha@gmail.com
1005	Anurag Lad	BCA	21	anurag@gmail.com

```
4 rows in set (0.00 sec)
```



```
mysql>
```


